

Dang Nguyen

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Research interests

My research focuses on data-centric methods for building efficient and reliable agentic AI systems based on large (vision-)language models. I develop techniques for **synthetic data generation** and **data selection** to construct high-quality training signals under limited or noisy data. More recently, I have explored reasoning and decision-making, including **test-time scaling**, **RL training**, and **uncertainty-aware inference**, with the goal of enabling **reliable multi-step agent** behavior in high-stakes settings.

Education

University of California, Los Angeles

Ph.D. in Computer Science

California, USA

Sep. 2023 - Present

- Advisor: Professor Baharan Mirzasoleiman
- UCLA Graduate Dean's Scholar Award

Toyo University

B.S. in Information Networking for Innovation and Design

Tokyo, Japan

Apr. 2017 - Mar. 2021

- Toyo Top Global Scholarship A
- GPA: 4.27/4.3, Top 1/300 in the faculty
- Top thesis award

Experience

Google Research

Student Researcher

California, USA

Sep. 2024 - Aug. 2025

- Topics: Synthetic data generation for LLMs and Data selection for LVLMs

Cisco

PhD Research Intern

California, USA

Jun. 2024 - Sep. 2024

- Supervisor: Dr. Ali Payani
- Topic: LLM Hallucination

VinAI (now Qualcomm AI)

AI Resident

Hanoi, Vietnam

Oct. 2020 - Aug. 2023

- Supervisor: Professor Nhat Ho (UT Austin)
- Topics: Optimal Transport and Model Merging
- Participated in an applied project which aims to improve the performance of object detectors in low-light conditions.
- Managed GPU resources for the VinAI Residency Program.

FPT Japan Holdings

Part-time Machine Learning Engineer

Yokohama, Japan

Oct. 2019 - Sep. 2020

- Participated in a long-term demand forecasting project for a chain pharmacy company in Japan.

Publications

(*) denotes equal contribution

1. N. Naharas*, **D. Nguyen***, N. Bulut, M. Bateni, V. Mirrokni, and B. Mirzasoleiman. Data Selection for Fine-tuning Vision Language Models via Cross Modal Alignment Trajectories. *International Conference on Machine Learning (ICML)*, 2026.
2. **D. Nguyen***, J. Li*, J. Zheng*, B. Mirzasoleiman. Do We Need All the Synthetic Data? Towards Targeted Synthetic Image Augmentation via Diffusion Models. *International Conference on Learning Representations (ICLR)*, 2026.
3. **D. Nguyen**, A. Payani, B. Mirzasoleiman. Beyond Semantic Entropy: Boosting LLM Uncertainty Quantification with Pairwise Semantic Similarity. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Findings) 2025.
4. **D. Nguyen***, Z. Li*, M. Bateni, V. Mirrokni, M. Razaviyayn, and B. Mirzasoleiman. Synthetic Text Generation for Training Large Language Models via Gradient Matching. *International Conference on Machine Learning (ICML)*, 2025.
5. **D. Nguyen**, W. Yang, R. Anand, Y. Yang and B. Mirzasoleiman. Mini-batch Coresets for Memory-efficient Language Model Training on Data Mixtures. *International Conference on Learning Representations (ICLR)*, 2025.
6. **D. Nguyen**, P. Haddad, E. Gan, and B. Mirzasoleiman. Changing the Training Data Distribution to Reduce Simplicity Bias Improves In-distribution Generalization. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
7. Y. Xue, J. Siddharth, **D. Nguyen**, and B. Mirzasoleiman. Understanding the Robustness of Multi-modal Contrastive Learning to Distribution Shift. *International Conference on Learning Representations (ICLR)*, 2024.
8. K. Nguyen*, **D. Nguyen***, N. Ho. Self-Attention Amortized Distributional Projection Optimization for Sliced Wasserstein Point-Cloud Reconstruction. *International Conference on Machine Learning (ICML)*, 2023.
9. **D. Nguyen**, T. Nguyen, K. Nguyen, D. Phung, H. Bui, and N. Ho. On cross-layer alignment for model fusion of heterogeneous neural networks. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023. (Top 3%)
10. K. Nguyen*, **D. Nguyen***, T. A. V. Le, T. Pham, and N. Ho. Improving mini-batch optimal transport via partial transportation. *International Conference on Machine Learning (ICML)*, 2022.
11. K. Nguyen, **D. Nguyen**, Q. Nguyen, T. Pham, H. Bui, D. Phung, T. Le, and N. Ho. On transportation of mini-batches: A hierarchical approach. *International Conference on Machine Learning (ICML)*, 2022.

Professional services

- Reviewer at Conference on Neural Information Processing Systems (NeurIPS) 2022-2026
- Reviewer at the International Conference on Machine Learning (ICML) 2023-2026
- Reviewer at the International Conference on Learning Representations (ICLR) 2024-2026
- Reviewer at the International Conference on Artificial Intelligence and Statistics (AISTATS) 2023-2025
- Program Committee at AAAI 2025
- Program Committee at New Frontiers in AdvML@NeurIPS2024
- Reviewer at Workshop on Spurious Correlation and Shortcut Learning @ ICLR 2025

Honors & Awards

INTERNATIONAL

2023	UCLA Graduate Dean's Scholar Award , UCLA	California, USA
2017	Toyo Top Global Scholarship A , Toyo University	Tokyo, Japan
2015	Silver medal , 56th International Mathematical Olympiad	Chiang Mai, Thailand

DOMESTIC

2015	First Prize , Vietnam Mathematical Olympiad	Hanoi, Vietnam
2014	Second Prize , Vietnam Mathematical Olympiad	Hanoi, Vietnam